

Jiri Crispeyn

Staff Software Engineer

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Experience

Shopify

Staff Software Engineer, June 2022 – Present

Technical leader across Shopify Point of Sale (POS) core infrastructure, performance architecture, and retail platform capabilities.

Technologies: React Native, TypeScript, Kotlin Multiplatform (KMM), Ruby on Rails, GraphQL (Apollo), SQLite, MySQL, Redux, Shopify Polaris, POS UI Extensions, Reanimated, Jest, Grafana, Looker, Stripe, Cursor/AI Tools

Retail Merchandising, May 2025 – Present

- Cart Performance Overhaul: Authored architectural blueprint, leading a complete performance refactor of the React Native POS cart domain. Eliminating redundant re-renders, overfetching, and DOM-tree inefficiencies, leveraging FlashList and smart memoization, cut time-to-render by 50% and solved scale degradation for large carts.
- Extensibility & Security: Drove POS support for Shopify's Cart & Checkout Validation Function API (pioneered via an internal Hackdays initiative), enabling offline-resilient, secure validation for 3P developers.
- Subscriptions & Vaulting: Partnered with Stripe and cross-functional payments teams to bring subscription/ selling plan capabilities to POS, introducing native card-vaulting infrastructure for recurring retail billing.
- Real-time Sync: Engineered concurrent, multi-device cart sharing backed by an in-house synchronization system supporting seamless offline-to-online state reconciliations.
- Discount System Migration: Directed zero-downtime migration of POS discount allocations, ensuring backwards compatibility and zero financial misallocations across complex checkout workflows.
- AI & Developer Productivity: Drove team-wide adoption of modern AI-assisted engineering workflows (Cursor, Claude Code, custom Model Context Protocol tools, and AI agents).

Retail Selling Strategies, May 2023 – May 2025

- Product Bundles Architecture: Technical lead for bringing Static and Dynamic Bundles to Shopify POS. Solved complex offline SQLite constraints by optimizing local DB schemas and background component fetching, driving a massive increase in retail GMV.
- Extensibility & Functions: Introduced Shopify Functions and the Cart Transform API to POS, building foundational POCs and opening dynamic bundling capabilities to third-party POS app developers.
- Cart Features & Catalogs: Created "Split Line Items" (via custom cart attributes) to enable itemized modifications (engraving, custom sizing) for retail merchants. Served as the primary retail point-of-contact for core Catalogs and B2B location-based pricing.
- Org-wide Engineering Quality: Championed retail-wide performance and polish sprints, eliminated tech debt (large-scale linting, barrel file removals, strict TypeScript typing), and served on staff advisory groups to enforce mobile performance standards.
- Mentorship & Leadership: Actively mentored engineers across the organization, successfully coaching peers through promotions from Software Engineer to Senior and Staff levels.

Retail Market Expansion, June 2022 – May 2023

- Compliance Platform Modularization: Re-architected the tightly coupled POS Retail Compliance module into a pluggable, modular platform. Built the foundation that enabled POS UI Extensions for third-party tax integrators and facilitated rapid international expansion.
- High-Resiliency Architecture: Designed offline-first tax reporting with React Native, Kotlin Multiplatform (KMM) coroutines, and local SQLite persistence, paired with sharded MySQL/Rails backends, achieving 99.9% tax reporting compliance accuracy.

- Observability & Failsafes: Built comprehensive Grafana monitoring, custom alert pipelines, and Sidekiq background processors to track real-time failsafes and system health.
- Unverified Returns Foundation: Authored the initial technical design for Unverified Returns, extending Shopify's core Order and Return relational models to support seamless in-store retail processing.

Aaqua

December 2020 – June 2022

Global social community platform connecting creators, brands, and fans.

Technologies: React Native, TypeScript, GraphQL (Apollo Federation), Nx Monorepo, AWS (Amplify, Pinpoint, IVS), Swift, Objective-C, Java, Storybook, Detox, Fastlane, CircleCI

Engineering Lead Mobile, December 2020 – June 2022

- Led mobile architecture and infrastructure for a large-scale React Native application, guiding engineering standards across cross-functional core teams.
- Drove the migration to an Nx monorepo, optimizing codebase structure, developer experience, and reducing CI pipeline costs by >50%.
- Directed a ground-up overhaul of the mobile design system; delivered a maintainable, System UI–inspired component library using Storybook and Styled Components.
- Overhauled the data fetching strategy by implementing Apollo Federation with co-located GraphQL fragments to optimize screen render performance and data isolation.
- Co-engineered the transition from an open-source video streaming solution to native player implementations (AVPlayer, ExoPlayer, and Amazon IVS).

Senior Software Engineer, December 2020 – June 2022

- Architected the foundational React Native codebase, establishing core services around React Navigation, AWS Amplify, and custom event tracking via AWS Pinpoint.
- Replaced Appium with Detox for end-to-end testing, drastically reducing test flakiness and improving execution speed.
- Automated build and release delivery using Fastlane and CircleCI with advanced caching and parallel test workflows.
- Established team engineering practices for advanced TypeScript patterns, testing standards with Jest, and hosted monthly Community of Practice knowledge-sharing sessions.

Sentiance

January 2016 – December 2020

Data science and behavioral AI company transforming sensor and motion data into contextual insights.

Technologies: TypeScript, React Native, Vue.js, Node.js, GraphQL, PostgreSQL, D3.js, Jest, Objective-C, Java

Senior Full Stack Engineer, January 2018 – October 2020

- Led a cross-functional team of engineers; owned engineering standards, technical hiring, code quality, and developer coaching.
- Engineered a white-label React Native app and Node.js/GraphQL backend that reduced distracted driving by ~35% across multiple client deployments.
- Re-architected legacy mobile infrastructure by migrating an Ionic 1 app to React Native, maintaining >96% test coverage.
- Consolidated three legacy AngularJS applications into a single real-time data visualization platform built with Vue.js.
- Authored and maintained open-source native bridges (Objective-C/Java) connecting the core Sentiance mobile SDK to JavaScript.

Full Stack Engineer, January 2016 – January 2018

- Developed Journeys, a showcase mobile app demonstrating real-time contextual sensor data, behavioral profiles, and timeline visualizations.
- Built real-time analytics dashboards using TypeScript, D3.js, Angular, RxJS, and GraphQL.

Nascom (acquired by Dynamate)

Front-End Engineer, March 2014 – January 2016

Technologies: Node.js, Express, JavaScript, Symfony, Drupal, Sass, REST APIs

- Built scalable SPAs, web applications, and enterprise CMS platforms for diverse client projects.
- Architected a high-concurrency Node.js queuing engine for high-demand ticket sales during a major global music festival.

Education

PXL University College

Bachelor in Computer Science (Cum Laude), 2011 – 2014

Skills

Mobile & Front-End

React Native, TypeScript, React, Remix, Vue.js, Reanimated, Redux (Saga, Toolkit), React Navigation, React Native Gesture Handler, Expo, TurboModules, D3.js, RxJS, Styled Components, Shopify Polaris, Shopify Skia

Native & Platform

Kotlin Multiplatform (KMM), Kotlin, Java, Swift, Objective-C

Back-End & API

Node.js, GraphQL (Apollo Federation), Ruby on Rails, PostgreSQL, SQLite, MySQL, Neo4J, WebSockets

Infrastructure & CI/CD

AWS (Amplify, Pinpoint, IVS), Firebase, Fastlane, CircleCI, Jenkins, Nx Monorepo, Git

Observability, AI & Tooling

Grafana, Datadog, Sentry, Bugsnag, LaunchDarkly, Jest, Detox, Shopify, Cursor, AI Agents (MCPs), pi.dev